Chapter 3. Resampling: Cross-Validation and the Bootstrap

Lecture 5 (validation set, LOOCV, K-fold CV, right and wrong CV, bootstrap SE and CI)

Contents

1 What Are Resampling Methods? 2

2 Cross-Validation 2

2.1 Validation-set approach . . . 2
2.2 Leave-one-out cross-validation . . . 2
2.3 K-fold cross-validation . . . 2
2.4 K-fold CV versus LOOCV . . . 3
2.5 Cross-validation: right and wrong . . . 3
2.6 What does CV estimate? . . . 3

3 The Bootstrap 3

3.1 A motivating example . . . 3
3.2 Back to the real world . . . 4
3.3 Bootstrap in general . . . 4
3.4 Confidence intervals . . . 4
3.5 The wrong way: bootstrap prediction error . . . 5

4 Selected exercises 5

1 What Are Resampling Methods?

Refit the model on repeated samples from the training set for

- **model assessment**: estimate test error rates;
- **model selection**: choose the level of flexibility.

Without a large test set, AIC/BIC adjust the training error (Ch. 2); here we hold out observations instead.

2 Cross-Validation

2.1 Validation-set approach

Randomly split into training and validation sets; the validation error (MSE, or misclassification rate) estimates the test error.

Limitations. (i) Highly variable, depending on the split.

(ii) Fitting on part of the data *overestimates* the test error of the full-data fit.

2.2 Leave-one-out cross-validation

Hold out one observation, fit on $n-1$, compute $\mathrm{MSE}_i = (y_i - \hat{y}_{(i)})^2$; repeat n times:

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{MSE}_i.$$

Less bias than the validation set, no split randomness, but n fits.

Theorem: LOOCV shortcut for least squares and linear smoothers

If $\hat{y} = Hy$ for a fixed matrix H with the leave-one-out property (true for least squares, ridge, smoothing splines), then

$$\mathrm{CV}_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2,$$

so LOOCV costs a single fit.

Proof for least squares [Supplement]. By the Sherman–Morrison formula, removing row i changes the estimate by $\hat{\beta} - \hat{\beta}_{(i)} = (X^\top X)^{-1} x_i e_i/(1-h_{ii})$ with $e_i = y_i - \hat{y}_i$. Hence $y_i - x_i^\top \hat{\beta}_{(i)} = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$. ■

High-leverage residuals are inflated.

2.3 K-fold cross-validation

Divide the data into K roughly equal parts $C_1, \ldots, C_K$ of sizes n_k; fit without part k and predict it:

$$\mathrm{CV}_{(K)} = \sum_{k=1}^{K}\frac{n_k}{n}\mathrm{MSE}_k, \qquad \mathrm{MSE}_k = \frac{1}{n_k}\sum_{i\in C_k}(y_i - \hat{y}_i)^2,$$

$\hat{y}_i$ fitted without part k; $K = n$ gives LOOCV.

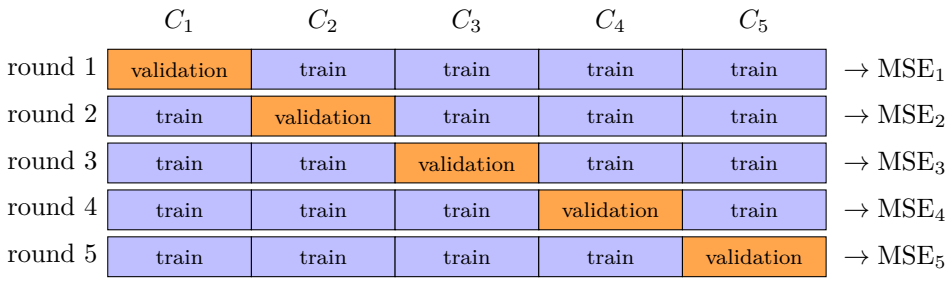


Figure 1: 5-fold cross-validation; each part is held out once.

2.4 K-fold CV versus LOOCV

Typically $K = 5$ or 10. Besides computation: the n LOOCV fits use nearly identical data, so their errors are *highly correlated* and the average has high variance. K-fold CV has more bias (training size $(K-1)n/K$) but lower variance. On the Auto data, LOOCV and nine repetitions of 10-fold CV give very similar curves.

2.5 Cross-validation: right and wrong

Example: selecting predictors before CV

5000 predictors, 200 samples:
(1) select the 100 predictors most correlated with the response;
(2) fit a model on these 100. CV on step 2 only? **No**: step 1 used all responses, so even for pure-noise predictors the CV error is near 0 while the true error is 50%. Right: in each fold, do steps 1 and 2 on the training part only.

Feature selection is training and must be inside the CV loop.

2.6 What does CV estimate?

CV estimates the prediction risk $R(\hat{f}) = \mathbb{E}[(Y - \hat{f}(X))^2]$.

Theory and limitation. LOOCV is asymptotically equivalent to AIC (Stone, 1977), but may prefer overly complex models, i.e. is not consistent for selection (Shao, 1997). Prediction → CV, AIC; true-model identification → BIC.

3 The Bootstrap

A tool to quantify the uncertainty of an estimator, e.g. a standard error or a confidence interval.

3.1 A motivating example

Invest α in an asset with return X and $1-\alpha$ in one with return Y; minimize $\text{Var}(\alpha X + (1-\alpha)Y)$.

Theorem: minimum-variance allocation

With $\sigma_X^2 = \text{Var}X$, $\sigma_Y^2 = \text{Var}Y$, $\sigma_{XY} = \text{Cov}(X, Y)$,

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof [Supplement]. $V(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$ is a convex quadratic; $V'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0$ gives $\alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}$. ■

Plug in estimates:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}.$$

Simulation (true $\alpha = 0.6$, 100 pairs): four data sets gave $\hat{\alpha} = 0.576, 0.532, 0.657, 0.651$; 1,000 repetitions gave $\bar{\alpha} = 0.5996$ and $\sqrt{\frac{1}{999}\sum_r(\hat{\alpha}_r - \bar{\alpha})^2} = 0.083 \approx \mathrm{SE}(\hat{\alpha})$.

3.2 Back to the real world

New population samples are unavailable; the bootstrap samples n observations **from the original data with replacement**. Histograms of 1,000 true-resampling and 1,000 bootstrap estimates look very similar.

Algorithm: bootstrap standard error

For $b = 1, \ldots, B$ (say $B = 100$ or 1000): draw Z^{*b} by sampling n observations with replacement, and compute $\hat{\alpha}^{*b}$. Then

$$\mathrm{SE}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}\right)^2}$$

estimates the standard error of $\hat{\alpha}$ computed from the original data.

Why it works. *Plug-in*: replace P_0 by $\hat{P}_n$ (mass $1/n$ per observation); for smooth statistics the law of $\hat{\alpha}^* - \hat{\alpha}$ approximates that of $\hat{\alpha} - \alpha$.

3.3 Bootstrap in general

For time series, the **block bootstrap** resamples blocks of consecutive observations, preserving short-range dependence.

3.4 Confidence intervals

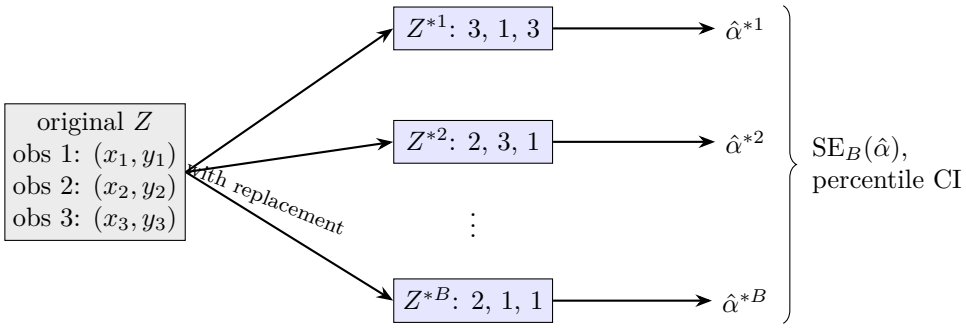


Figure 2: The bootstrap with $n = 3$.

The 5% and 95% quantiles of 1,000 bootstrap $\hat{\alpha}^*$ are $(0.43, 0.72)$: an approximate 90% **bootstrap percentile interval**. Alternatives: normal $\hat{\alpha} \pm z_{1-\alpha/2}\mathrm{SE}_B$; basic $(2\hat{\alpha} - q_{0.95},\ 2\hat{\alpha} - q_{0.05})$.

3.5 The wrong way: bootstrap prediction error

Training on bootstrap samples and validating on the original sample: an observation is in a bootstrap sample with probability $1-(1-1/n)^n \approx 0.632$ (Exercise 3.1), so the prediction error is seriously *underestimated*. Out-of-bag observations fix this (".632" estimators, OOB error in bagging, Ch. 11).

Summary

- Validation set: variable, biased upward; LOOCV: nearly unbiased, highly correlated fold errors; $K = 5$ or 10 balances both.
- Every data-dependent step (including variable selection) must be inside the CV loop.
- Prediction → cross-validation (like AIC); uncertainty → bootstrap.

4 Selected exercises

Motivation. The 0.632 inclusion probability (ISLR Ex. 5.2), the ridge LOOCV shortcut, and the exact bootstrap SE of a mean.

Exercise 3.1 (Inclusion in a bootstrap sample; ISLR Ex. 5.2)

A bootstrap sample draws n observations with replacement from n observations.
(a) What is the probability that the first bootstrap observation is not the jth original observation? The second?
(b) Show that the probability that observation j is not in the bootstrap sample is $(1-1/n)^n$, and find the expected number of distinct original observations in a bootstrap sample.
(c) Evaluate the inclusion probability for $n = 5, 100, 10\,000$, and prove that it decreases strictly to $1-e^{-1}$ as $n \to \infty$ ($n \geq 2$).

Solution.

(a) Each draw is uniform on $\{1,\dots,n\}$, so the first draw differs from j with probability $1-1/n$; the same holds for the second draw.

(b) The n draws are independent, so $\Pr(j \notin \text{sample}) = (1-1/n)^n$. With I_j the indicator that j appears, the number of distinct observations is $\sum_j I_j$, with expectation $n\{1-(1-1/n)^n\}$ by linearity.

(c) The values $1-(1-1/n)^n$ are 0.672, 0.634 and 0.632.

Step 1. Write $(1-1/n)^n = e^{a_n}$ with $a_n = \varphi(1/n)$, where $\varphi(x) = \log(1-x)/x$ for $0 < x < 1$. From $\log(1-x) = -\sum_{k\geq 1} x^k/k$,

$$\varphi(x) = -\Big(1 + \frac{x}{2} + \frac{x^2}{3} + \cdots\Big),$$

which is strictly decreasing in x with $\varphi(x) \to -1$ as $x \downarrow 0$.

Step 2. As n increases, $x = 1/n$ decreases, so $a_n \uparrow -1$ strictly and $(1-1/n)^n \uparrow e^{-1}$ strictly; the inclusion probability decreases strictly to $1-e^{-1} \approx 0.632$. ■

Exercise 3.2 (LOOCV shortcut for ridge regression)

Let $X \in \mathbb{R}^{n\times p}$ (rows $x_i^\top$, no intercept), $\lambda > 0$, $A = X^\top X + \lambda I$, $\hat\beta = A^{-1}X^\top y$, $\hat y = X\hat\beta$, and $h_{ii} = x_i^\top A^{-1} x_i$. Let $\hat\beta_{(i)}$ be the ridge estimate with the same λ computed without observation

i.
(a) Prove the Sherman–Morrison formula: if A is invertible and $c = 1 - u^\top A^{-1}u \neq 0$, then $(A - uu^\top)^{-1} = A^{-1} + A^{-1}uu^\top A^{-1}/c$.
(b) Show that $h_{ii} < 1$ and $y_i - x_i^\top \hat\beta_{(i)} = (y_i - \hat y_i)/(1 - h_{ii})$.

Solution [Supplement].

(a) Multiplying out,

$$(A - uu^\top)\Big(A^{-1} + \frac{A^{-1}uu^\top A^{-1}}{c}\Big) = I + \frac{uu^\top A^{-1}}{c} - uu^\top A^{-1} - \frac{u(u^\top A^{-1}u)u^\top A^{-1}}{c}$$
$$= I + uu^\top A^{-1}\,\frac{1 - c - u^\top A^{-1}u}{c} = I.$$

(b)

Step 1 ($h_{ii} < 1$). If $x_i = 0$ then $h_{ii} = 0$. Otherwise put $v = A^{-1}x_i \neq 0$. For every w, $w^\top Aw = \|Xw\|^2 + \lambda\|w\|^2 \geq (x_i^\top w)^2 + \lambda\|w\|^2$. With $w = v$: $v^\top Av = x_i^\top A^{-1}x_i = h_{ii}$ and $x_i^\top v = h_{ii}$, so $h_{ii} \geq h_{ii}^2 + \lambda\|v\|^2 > h_{ii}^2$, i.e. $0 \leq h_{ii} < 1$.

Step 2. Deleting row i gives $A_{(i)} = A - x_ix_i^\top$ and $\hat\beta_{(i)} = A_{(i)}^{-1}(X^\top y - x_iy_i)$. By (a) with $u = x_i$ and $c = 1 - h_{ii} > 0$,

$$x_i^\top A_{(i)}^{-1} = x_i^\top A^{-1} + \frac{h_{ii}\,x_i^\top A^{-1}}{1 - h_{ii}} = \frac{x_i^\top A^{-1}}{1 - h_{ii}}.$$

Step 3. Therefore $x_i^\top \hat\beta_{(i)} = \dfrac{x_i^\top A^{-1}X^\top y - h_{ii}y_i}{1 - h_{ii}} = \dfrac{\hat y_i - h_{ii}y_i}{1 - h_{ii}}$, and

$$y_i - x_i^\top \hat\beta_{(i)} = \frac{(1 - h_{ii})y_i - \hat y_i + h_{ii}y_i}{1 - h_{ii}} = \frac{y_i - \hat y_i}{1 - h_{ii}}.$$

$\lambda = 0$ is the least-squares case, provided $A_{(i)}$ is invertible. (With re-centering/re-standardizing inside each fold the identity is only approximate.) ■

Exercise 3.3 (Bootstrap standard error of the mean)

Fix data $x_1, \ldots, x_n$ and let $x_1^*, \ldots, x_n^*$ be drawn i.i.d. from the empirical distribution $\hat P_n$, with mean $\bar x^*$.
(a) Show that $\mathbb{E}^*\bar x^* = \bar x$ and $\mathrm{Var}^*\bar x^* = \hat\sigma^2/n$, where $\hat\sigma^2 = \frac{1}{n}\sum_i (x_i - \bar x)^2$.
(b) Show that $\mathrm{SE}_B \to \hat\sigma/\sqrt{n} = \sqrt{(n-1)/n}\; s/\sqrt{n}$ almost surely as $B \to \infty$, with s^2 the usual sample variance.

Solution [Supplement].

(a) Each x_j^* takes the value x_i with probability $1/n$, so $\mathbb{E}^*x_j^* = \bar x$ and $\mathrm{Var}^*x_j^* = \frac{1}{n}\sum_i (x_i - \bar x)^2 = \hat\sigma^2$. The mean of n i.i.d. copies has mean $\bar x$ and variance $\hat\sigma^2/n$.

(b) Conditionally on the data, the bootstrap replicates $\bar x^{*1}, \ldots, \bar x^{*B}$ are i.i.d. and bounded. By the strong law of large numbers, $\frac{1}{B}\sum_r \bar x^{*r} \to \bar x$ and $\frac{1}{B}\sum_r (\bar x^{*r})^2 \to \hat\sigma^2/n + \bar x^2$ almost surely, so

$$\mathrm{SE}_B^2 = \frac{B}{B-1}\Big[\frac{1}{B}\sum_r (\bar x^{*r})^2 - \Big(\frac{1}{B}\sum_r \bar x^{*r}\Big)^2\Big] \to \frac{\hat\sigma^2}{n}.$$

Since $\hat\sigma^2 = (n-1)s^2/n$, the limit is $\sqrt{(n-1)/n}\; s/\sqrt{n}$. ■